That's a fantastic idea — building an end-to-end **agentic AI model for digital forensics** could really showcase how AI can augment traditional forensic tools like **Autopsy**. Let's break down how you might structure your proof of concept:

🔍 Core Components of Your AI-Driven Forensics Model

1. Data Ingestion Layer

- **Sources:** Disk images, memory dumps, logs, network captures.
- **Tools:** Autopsy (for disk image parsing), Volatility (for memory analysis), Wireshark/tshark (for network traffic).
- **AI Role:** Automate preprocessing — e.g., classifying file types, extracting metadata, anomaly detection.

2. Forensic Processing Pipeline

- **Autopsy Integration:** Use Autopsy modules for file system analysis, keyword search, timeline building.
- **AI Agent Tasks:**
  - Automate evidence triage (flagging suspicious files, hidden partitions).
  - Apply NLP models to detect sensitive or incriminating text in documents/emails.
  - Use ML anomaly detection for unusual system events.

3. Knowledge Graph / Case Context

- Build a **graph database (Neo4j)** linking artifacts:
  - Files → Users → Timestamps → Network activity.
- AI agent can query this graph to generate hypotheses (e.g., "User X accessed File Y before deletion").

4. Decision-Making Agent

- **Agentic AI Layer:** Orchestrates tools and reasoning.
  - Example: If suspicious executable found → run YARA rules → cross-check with VirusTotal API.
- Could use frameworks like **LangChain** or **Semantic Kernel** to chain forensic tasks.

5. Reporting & Visualization

- AI generates **structured forensic reports**:
  - Timeline of events
  - Key evidence summaries
  - Confidence scores for findings
- Visualization with tools like Kibana or custom dashboards.

⚙️ Suggested Proof-of-Concept Workflow

1. **Input:** A disk image from a compromised system.
2. **Autopsy:** Extract file system artifacts.
3. **AI Agent:**
   a. Classify files (benign vs suspicious).
   b. Summarize user activity.
   c. Detect anomalies (e.g., deleted logs, hidden files).
4. **Graph Database:** Link evidence into a timeline.
5. **Output:** Automated forensic report with AI-driven insights.

🚀 Next Steps

- Start small: integrate **Autopsy + AI text classification** for document analysis.
- Expand: add **Volatility** for memory forensics, then build orchestration logic.
- Long-term: create a **multi-agent system** where each agent specializes (disk, memory, network, reporting).

Would you like me to sketch out a **reference architecture diagram** for this AI-forensics pipeline so you can visualize how the agents, tools, and processes connect?